# MODULE 0 — What “Data Science” Means to AI

**Terminology**

- **SAP BW tools -** refers to **SAP Business Warehouse,** an enterprise data warehousing platform used by large organizations to collect, transform, & report on data from SAP & non-SAP systems.
  - In practice usually means the ETL, data modeling, & reporting components inside SAP BW that support enterprise analytics & decision-making rather than ad hoc data science work.
- **Label delay -** means the outcome you care about (example: true customer churn) is only observed *after* a significant time lag, so recent data may be mislabeled as "not churned" simply because the event hasn't happened yet
- **Label noise -** means some recorded outcomes are wrong or ambiguous (e.g., customers marked as churned for administrative reasons), causing the system to learn patterns that reflect logging artifacts rather than real behavior.

## Why This Module Exists

Humans often define data science by **tools**. AI defines data science by **decision structure**.
If you don’t align on this early, everything downstream breaks:

- models optimize the wrong thing
- metrics don’t map to decisions
- production teams inherit undefendable systems

This module **reprograms your mental model** before you touch ingestion, cleaning, or modeling.

## Core Mental Model (Anchor This)

**To AI, data science is the practice of transforming uncertainty into constrained, defensible decisions using data.**
Pipelines are not scripts.They are **decision graphs**.

Takeaways

- Historically data science has been taught as "something you do" before delivering analysis (e.g., loading, cleaning, merging, etc.) -> 80% of DS work is data preparation, 20% is modeling (as the saying would go).
  - **Issue:** this mental model naturally narrows practitioners point of view from what matters to an organization (e.g., tangible meaning of the work) -> it's why conversations tend to become technical very quickly
    - Everything spoken about is tool & technique driven
    - How the work ties back to the organization is usually thought through at the very end.
  - **Issue:** this causes discipline to become blurry, practitioners find themselves involved in unnecessary technical deep dives that span across other disciplines (data engineering, database administration, IT, etc).
- DS's role is to transform uncertainty into something structured & explainable -> DS is not simply being a technician with a toolbelt.
  - It's not the stakeholder's responsibility to turn business needs into formulated projects
  - **Moving away** from "using this tool" to "do this thing" -> **moving towards addressing** "what are we trying to do?", "what does that need to look like?", "what would having this help us achieve?", "what would failure mean to us out of this?" -> which then drives what actions need to occur & those actions drive what tools & techniques will be required

## How AI Sees the Data Science Lifecycle

AI implicitly reasons *backwards*:

1. **What decision is being made?**
2. **What uncertainty affects that decision?**
3. **What evidence reduces that uncertainty?**
4. **What representation (features, summaries) makes that evidence usable?**
5. **Is a model even necessary?**

Most human workflows start at step 4 or 5. AI starts at step 1 — and everything else must justify itself.

## Key Reframes (Very Important)

### ❌ Old framing

- “I’m building a model to predict X”
- “This notebook does EDA and modeling”
- “We need better features”

### ✅ AI-native framing

- "We are deciding **X** under **Y constraints** with **Z uncertainty**"
- "This step reduces uncertainty about **[specific risk]**"
- "This representation changes what the model can see"

This language shift is subtle — but it's the difference between *analysis* and *decision support*.

## AI Decision Questions (You'll Reuse These Constantly)

At **every** pipeline stage, AI implicitly asks:

1. **What decision is downstream of this?**
2. **What uncertainty am I reducing?**
3. **What constraints matter most here?**

If you cannot answer all three, the step is likely misaligned or unnecessary.

## Unifying Idea of Data Science Across Lesson Plan

**A data science pipeline is a sequence of irreversible decisions that progressively collapse uncertainty into structure.**

Each step:

- answers some questions
- *locks in* some assumptions
- eliminates entire classes of future options

This is why early steps will feel abstract and later steps will feel constrained.

## 📊 Example Project Scenario (Shared Baseline)

### Project Idea: Customer Retention Intervention System

A subscription-based company has noticed that **customer churn has increased over the last two quarters**. Leadership wants to reduce churn, but interventions (discounts, outreach, service upgrades) are **costly and limited**, so they cannot be applied to everyone.

A data science team is asked to "analyze the data and build a model" to help address the issue.

Available data includes:

- customer usage history
- billing and tenure information
- support ticket volume
- past churn outcomes (with noise and delays)

There is uncertainty about:

- which signals actually precede churn
- whether churn is behavior-driven or price-driven
- how reliable historical labels are

## 🧪 Revised Module 0 Mini-Exercise (Decision-First)

### Step 1 — Rewrite the project goal as a decision statement

Use this template:

**"We are deciding *[action]* under *[constraints]* with uncertainty about *[unknowns]*."**

**Example solution (one of many valid answers):**

"We are deciding which customers should receive retention interventions under budget and operational constraints, with uncertainty about whether observed behavior indicates true churn risk or temporary noise."

### Step 2 — Sanity-check alignment

Ask:

- Who takes this action?
- What happens if the decision is wrong?
- Is this decision reversible?

If those answers are fuzzy, the *problem definition* — not the model — is the issue.

This scenario is intentionally generic, realistic, and reusable.We'll continue to reuse it lightly across **ingestion, cleaning, features, modeling, and evaluation** so you can see how the *same decision* propagates through the pipeline.

When you're ready, we'll move on to **Module 1 — Data Ingestion: How AI Thinks About Inputs**.

## How AI Should Be Used *Here* (Discipline Check)

At this stage, AI's role is:

- **translator** (turning tasks into decisions)
- **challenger** (questioning assumptions)
- **enumerator** (listing uncertainties)

AI should **not**:

- design full pipelines yet
- suggest models

- optimize metrics

We'll earn that later.

## What You've Gained After Module 0

By the end of this module, you can:

- explain data science without naming tools
- frame work in AI-native decision language
- catch misaligned pipelines *before* they harden
- sound production-aware without touching production

This is the **lowest-risk, highest-signal** shift you can make at work.

Takeaways

- AI-native nuance when it comes to stating whether a decision is reversible or not
  - 💸 **Cost is irreversible**
  - 📉 **Customer churn outcomes may be irreversible (partial)**
  - 🔁 **Policy decisions *can* be revised in future cycles**

## Relevant Articles

*Basic Introduction to Data Science Pipeline*

- https://www.analyticsvidhya.com/blog/2022/08/basic-introduction-to-data-science-pipeline/?utm_source=chatgpt.com

*Mastering Decision Making and Uncertainty: A Leader's Guide*

- https://www.sapbwconsulting.com/blog/decision-making-and-uncertainty?utm_source=chatgpt.com

*Role of Data Science in Informed Business Decision Making*

- https://codiant.com/blog/role-of-data-science-in-informed-business-decision-making/?utm_source=chatgpt.com

*Decision Tree in Machine Learning*

- https://www.geeksforgeeks.org/machine-learning/decision-tree-introduction-example/?utm_source=chatgpt.com

*Data Analytics Pipeline 2026: Tools, Techniques, and Effective Strategies*

- https://kanerika.com/blogs/data-analytics-pipeline/?utm_source=chatgpt.com

*Data Pipeline Architecture: Components & Diagrams Explained*

- https://www.timeplus.com/post/data-pipeline-architecture?utm_source=chatgpt.com